

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY CHANGA

Third Semester of B. Tech. (Backlog) Examination (CL)
May 2011

CL 203 Fluid Mechanics

Date: 19.05.2011, Thursday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Define: Density, Specific Gravity, Atmospheric Pressure and Absolute Pressure [4]
(b) State Newton's Law of Viscosity. Discuss the control volume concept and fluid as a continuum in brief. [3]
- Q - 2 (a) Differentiate between U-tube and inverted U-tube differential manometer? [5]
(b) Define Total Pressure and Center of Pressure [2]
(c) Explain briefly: Mach Number, Mach Cone and Mach Angle [3]
(d) A horizontal venturimeter with inlet and throat diameter 50 cm and 30 cm, respectively is used to measure the flow of water. The readings of differential manometer connected to the inlet and throat is 18 cm of mercury. Determine the discharge. Take $C_d = 0.98$. [4]

OR

- Q - 2 (a) What are the gauge and absolute pressure at a point 5 m from the free surface of a liquid having density of liquid $1.25 \times 10^3 \text{ kg/m}^3$, if the atmospheric pressure is equivalent to 750 mm mercury? The specific gravity of mercury is 13.6 and density of water is 1000 kg/m^3 . [5]
(b) Define: Buoyancy, Center of Buoyancy, Compressible flow, Speed of sound wave and Shock wave [5]
(c) Find the value of oil-mercury differential manometer having $C_d = 0.98$, if discharge of oil flowing through the horizontal venturimeter is 40 lit/sec. The venturimeter has inlet and throat diameters 20 cm and 10 cm, respectively. Take specific gravity of oil = 0.9. [4]
- Q - 3 (a) Calculate Mach number at a point on a jet propelled aircraft, which is flying at 1120 km/hour at sea-level where air temperature is 25°C . [3]
Take, $k = 1.4$ and $R = 287 \text{ J/kg K}$.

- (b) Briefly discuss the following terms: [4]
 (i) Boundary Layer Thickness, δ (ii) Displacement Thickness, δ^*
- (c) Differentiate laminar flow and turbulent flow. [4]
- (d) Write the Bernoulli's equation and state assumptions made for such a derivation. [3]

OR

- Q - 3 (a) A projectile is traveling in air having pressure and temperature as 9.9 N/cm^2 and $(-4)^\circ\text{C}$. If the Mach angle is 43° , find the velocity of the projectile. Take $k = 1.4$ and $R = 287 \text{ J/kg K}$. [3]
- (b) Derive the equation for viscous fluid flow between two parallel plates. [4]
- (c) What power is required per kilometer of a line to overcome the viscous resistance to the flow of glycerin through a horizontal pipe of diameter 120 mm at the rate of 12lit/sec? Take, $\mu = 8$ poise and kinematic viscosity (ν) = 6 stokes. [4]
- (d) A pipe of diameter 400 mm carries water at a velocity of 25 m/sec. The pressures at the points 'A' and 'B' are given as 29.430 N/cm^2 and 22.563 N/cm^2 , respectively. The datum head at 'A' and 'B' are 28 m and 30 m. Find the loss of head between 'A' and 'B'. [3]

SECTION – II

- Q - 4 (a) Discuss the free vortex flow in detail. [4]
- (b) An open circular tank of 40 cm diameter and 150 cm long contains water up to a height of 70 cm. The tank is rotated about its vertical axis at 260 r.p.m. Find the depth of parabola formed at the free surface of water. [3]
- Q - 5 (a) For a two-dimensional potential flow, the stream function is given by $\psi = 2xy$, calculate the velocity at the point (3, 6). [5]
- (b) Discuss and derive equation for loss of head due to sudden-contraction in pipe flow. [5]
- (c) Calculate discharge through a rectangular orifice 2.0 m wide and 1.6 m deep fitted to a tank of water. The water level in the tank is 2.7 m above the top edge of the orifice. Take coefficient of discharge as 0.64. [4]

OR

- Q - 5 (a) Discuss Lagrangian method of motion of fluid particles. [5]
- (b) Discuss hydraulic gradient and total energy line with neat sketch. [5]
- (c) Calculate the discharge over rectangular notch of length 0.2 m and head over notch as 0.9 m. Take coefficient of discharge as 0.62. [4]
- Q - 6 (a) Describe procedure to conduct dimensional analysis using Buckingham's Π -Theorem. [5]

- (b) What are various dimensionless numbers? Explain any one in detail. [4]
- (c) Prove that the ratio of maximum velocity to average velocity is 2 for the viscous flow in a pipe. [5]

OR

- Q - 6 (a) Derive an expression for the viscous force 'F' by assuming that force 'F' exerted by a fluid on a sphere of diameter 'D' depends on the dynamic viscosity ' μ ', mass density of the fluid ' ρ ' and the velocity of sphere ' v '. [5]
- (b) Discuss dimensional homogeneity by giving suitable examples. [4]
- (c) A 2000 m long pipeline is used for transmission of 115 kW power. The pressure of water at inlet of pipe is 4 MPa. The pressure drop over the length of pipe is 0.8MPa. Calculate the efficiency of transmission. Take $f = 0.005$. [5]
